

The goal was to analyze model interpretability and fairness using the Titanic dataset.

```
!pip install shap
```

```
Requirement already satisfied: shap in /usr/local/lib/python3.12/dist-packages (0.52.0)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from shap) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from shap) (1.16.3)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from shap) (1.6.1)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from shap) (2.2.2)
Requirement already satisfied: tqdm>=4.27.0 in /usr/local/lib/python3.12/dist-packages (from shap) (4.67.3)
Requirement already satisfied: packaging>20.9 in /usr/local/lib/python3.12/dist-packages (from shap) (26.2)
Requirement already satisfied: slicer==0.0.8 in /usr/local/lib/python3.12/dist-packages (from shap) (0.0.8)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages (from shap) (0.60.0)
Requirement already satisfied: llvmlite in /usr/local/lib/python3.12/dist-packages (from shap) (0.43.0)
Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from shap) (3.1.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->shap) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->shap) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->shap) (2026.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->shap) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn->shap) (3.5.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->shap) (1.17.0)
```

```
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

df = pd.read_csv(url)

df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques	female	35.0	1	0	113803	53.1000	C123	C

```
data = df[['Pclass', 'Sex', 'Age', 'Fare', 'Survived']]

data = data.dropna()

data['Sex'] = data['Sex'].map({
    'male':0,
    'female':1
})

data.head()
```

	Pclass	Sex	Age	Fare	Survived
0	3	0	22.0	7.2500	0
1	1	1	38.0	71.2833	1
2	3	1	26.0	7.9250	1
3	1	1	35.0	53.1000	1
4	3	0	35.0	8.0500	0

```
from sklearn.model_selection import train_test_split

X = data.drop('Survived', axis=1)
y = data['Survived']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
from sklearn.ensemble import RandomForestClassifier

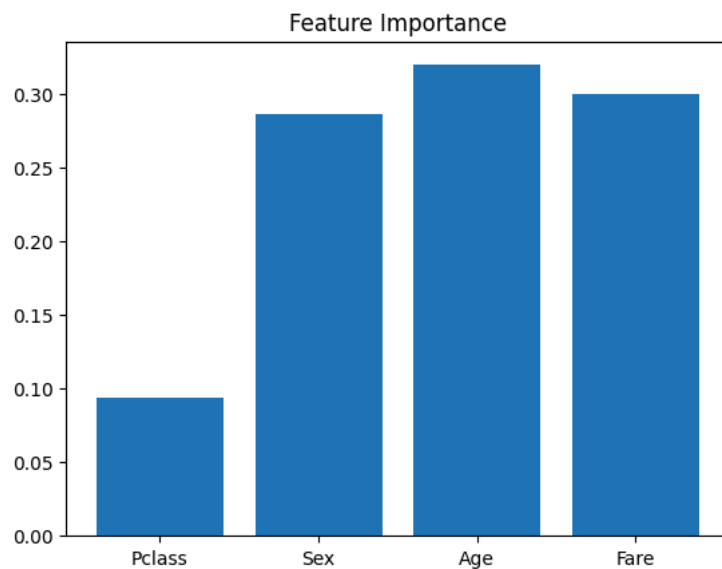
model = RandomForestClassifier()
```

```
model.fit(X_train, y_train)
```

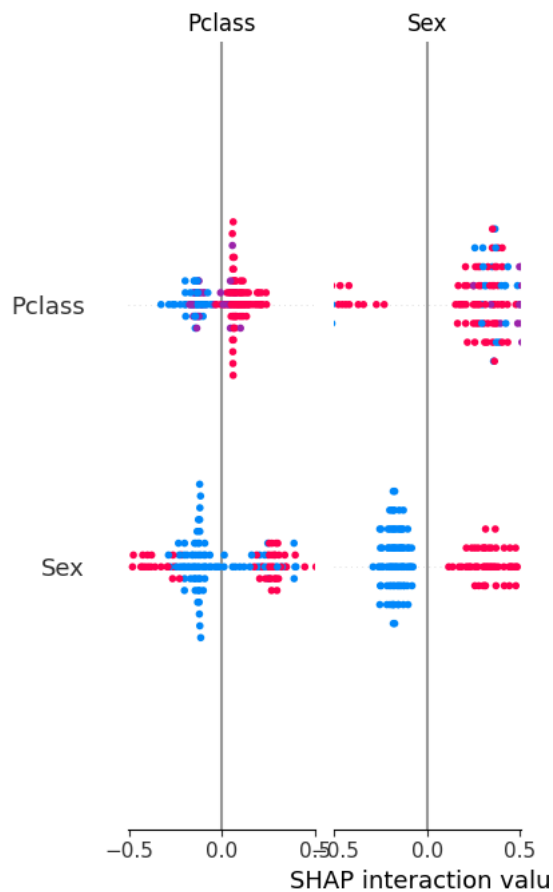
```
▼ RandomForestClassifier ⓘ ?  
RandomForestClassifier()
```

```
from sklearn.metrics import accuracy_score  
  
pred = model.predict(X_test)  
  
accuracy = accuracy_score(y_test, pred)  
  
print("Accuracy:", accuracy)  
  
Accuracy: 0.7762237762237763
```

```
import matplotlib.pyplot as plt  
  
importance = model.feature_importances_  
  
features = X.columns  
  
plt.bar(features, importance)  
  
plt.title("Feature Importance")  
plt.show()
```



```
import shap  
  
explainer = shap.TreeExplainer(model)  
  
shap_values = explainer.shap_values(X_test)  
  
shap.summary_plot(  
    shap_values,  
    X_test  
)
```

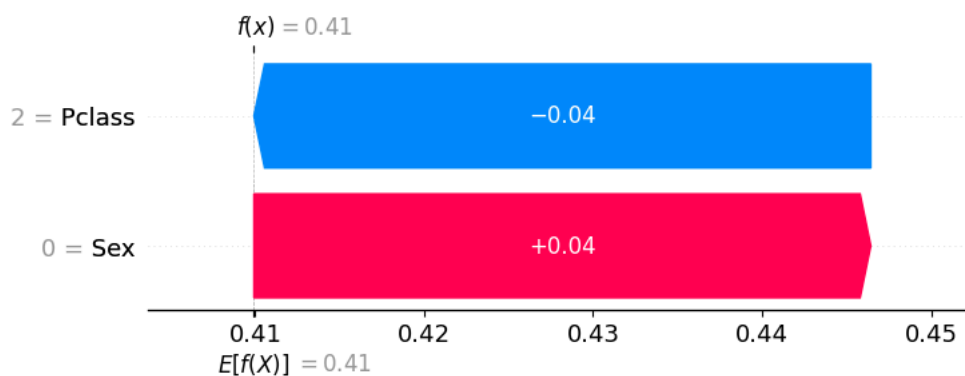


```
shap.force_plot(
    explainer.expected_value[1],
    shap_values[0][:, 1],
    X_test.iloc[0].values,
    feature_names=X_test.columns.tolist()
)
```

Visualization omitted, Javascript library not loaded!

Have you run 'initjs()' in this notebook? If this notebook was from another user you must also trust this notebook (File -> Trust notebook). If you are viewing this notebook on github the Javascript has been stripped for security. If you are using JupyterLab this error is because a JupyterLab extension has not yet been written.

```
shap.plots.waterfall(
    shap.Explanation(
        values=shap_values[1][0], # SHAP values for the first instance, class 1
        base_values=explainer.expected_value[1],
        data=X_test.iloc[0].values,
        feature_names=X_test.columns.tolist()
    )
)
```



```
male = X_test['Sex'] == 0
female = X_test['Sex'] == 1

from sklearn.metrics import accuracy_score

male_acc = accuracy_score(
    y_test[male],
```

```
        pred[male]
    )

    female_acc = accuracy_score(
        y_test[female],
        pred[female]
    )

    print("Male Accuracy:", male_acc)
    print("Female Accuracy:", female_acc)
```

```
Male Accuracy: 0.7752808988764045
Female Accuracy: 0.7777777777777778
```

conclusion

The model achieved good predictive performance and SHAP provided transparent explanations. Bias analysis highlighted the importance of evaluating fairness before deploying AI systems.